

ALGORITHM – MULTILEVEL INHERITANCE

Inheritance Structure: $A \rightarrow B \rightarrow C$

1. Start.
2. Create class A with variables **a = 100**, **b = 10** and method **sum()** to calculate $a + b$.
3. Create class B that extends A. Set **a = 10**, **b = 50** and create **sub()** to calculate $b - a$.
4. Create class C that extends B. Set **a = 25**, **b = 25** and create **mul()** to calculate $a \times b$.
5. In the main() method, create an object **tv** of class C.
6. Call **tv.sub()**. Calculate $50 - 10$ and display **40**.
7. Call **tv.mul()**. Calculate 25×25 and display **625**.
8. Call **tv.sum()**. Calculate $100 + 10$ and display **110**.
9. Display all the results.
10. Stop.

Quick Flow

START → Create A → Create B extends A → Create C extends B → Create C object → **sub()** → **mul()** → **sum()** → Display output → **STOP**

Output

subtraction is 40

multiplication is 625

sum is 110

Mnemonic: $A \rightarrow B \rightarrow C$ = Multilevel Inheritance